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(1) Research Program/Project/Study Title:	Developmental of Mung Beans Gelatin Dessert Plant-based Milk
Proponent Name:	Juvy I. Dordas
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	Acceptability of Mung Beans Gelatin Dessert
Component Leader:	none
Academic Rank:	none

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Agriculture/Fruit and Vegetables		(8) Discipline/Field of Specialization Food and Development							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The mung bean (*Vigna radiata* L.) is one of the most significant edible legume crops which accounts for approximately 6 million hectares of crop area in the Philippines (almost 8.5 percent of global pulse area) and consumed by most Asian families. San Mateo, Isabela is the Philippines' mung bean capital. Dioses et al. (2022)

The legumes are widely farmed in the Philippines and it is the most common and popular among Filipinos vegetable dish with pork, chicken and shrimps eaten by the family. According to Muinos (2021) like other types of legumes, mung beans are a rich source of plant-based protein, complex carbohydrates, fiber, and other nutrients.

According to Julio P. Yap Jr. of agriculture magazine the Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development of the Department of Science and Technology (DOST-PCAARRD) acting executive director Dr. Reynaldo V. Ebor, identified four major mung bean producing regions in the country are Region 2 (Cagayan Valley); Region 3 (Central Luzon); Region 11 (Davao Region) and Region 6 (Western Visayas).

Many researchers conducted and developed various products using mung beans such as flour, utilizing mung beans in cookie products, pastries, and even desserts. However, the researcher finds problems or gaps in those research study in which baking product are costly and needs extra effort and prior knowledge in baking. With this, the researcher comes up with ideas to make mung bean gelatin dessert that is affordable, easy to prepare for everyone, nutritious and profitable. (justify)

The significance of this research output is individual and community ideas to start and venture a small additional source of income to a family in a small budget and less effort as well, hoping slowly uplift their lives. The people in the community will be encouraged and motivated to cultivate mung beans so it would have an available source through this it would be an answer to current needs issues of unemployment.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The proposed project will be pass through trial and error testing or experimentation. The sensory attribute of the product will be evaluated by the semi-trained panelist during formulation of the product and the consumer testing will be at random. Both are uses (9) nine point hedonic scale instrument to validate the good quality of the product.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives: The general objective is to a healthy yet affordable gelatin dessert product with a twist on the common dessert which is the "Mung Beans Gelatin Dessert". Specifically, to obtain the following;

1. Describe the sensory qualities of the Mung Beans dessert gelatin plant-based milk in different proportions in terms of appearance, aroma, taste, and texture evaluated by the trained panelist.
2. Determine the Acceptability of the Mung Beans dessert gelatin plant-based milkin terms of appearance, aroma, taste, and texture evaluated by the consumer.
3. Find out if there is a significant difference in the sensory qualities of Mung Beans dessert gelatin plant-based milk in different treatments of proportion in terms of the appearance, aroma, taste, and texture
4. Find out if there is a significant difference in the acceptability of Mung Beans dessert gelatin plant-based milk in different treatments in terms of the appearance, aroma, taste, and texture
5. Determine the shelf-life of mung beans gelatin dessert gelatin plant-based milk at chilling temperature

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

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The present study will be based on the following prior arts such as:

Mung bean flour is commonly made into a paste and incorporated into bread and desserts according to Mendoza et al. (2001).

According to Noor Aziah,, Mohamad Noor, & Ho, (2012) This study was to investigate the effects of wheat flour substitution with legume flour (mung bean and chick pea) in cookies in terms of the physicochemical and organoleptic properties. Three formulations of cookies were prepared from (a) Control (100% wheat flour), (b) Mung bean (50% wheat flour + 35% mung bean flour + 15% corn flour) and (c) Chickpea (50% wheat flour + 35% chickpea flour + 15% corn flour). The physicochemical and organoleptic attributes of the three types of cookie were evaluated. Results showed significant different ($p<0.05$) in terms of ash, protein, crude fibre and total carbohydrate among cookies. Chickpea cookies was significantly ($p<0.05$) highest in protein and resistant starch content among the three types of cookies. The mung bean cookies was significantly ($p<0.05$) highest in weight, diameter, height and spread ratio. Textural measurement showed chickpeas cookies was significantly highest ($p<0.05$) in hardness, crispiness, elasticity, gumminess, and chewiness than the other two types of samples evaluated. For sensory evaluation, chickpeas cookies showed significant high difference in flavor , crispiness and aftertaste attributes but insignificant ($p>0.05$) different between mungbean cookies in term of overall acceptability. Chickpea cookies had the best flavour , crispiness and acceptability.

Stated by Mendoza et al. (1988) The Philippines produces about 33,000 mt of mung bean (*Vigna radiata* (L.) Wilczek) annually (BAECON, 1981-1983) with average yield of 0.651 5/ha from 50,000 ha. Mung bean bean is widely utilized as whole or germinated seeds and can be processed into flour or noodles. Thus, whole mung bean seeds are used in native delicacies such as "butse-butse" and "hopia", as a soup, or combined with sugar as snacks or dessert (PCARR, 1977). Germinated seeds or sprouts are mixed with shrimp and/or meat and used as vegetable dish. A portion of wheat flour can be substituted with mung bean flour to obtain high-protein bread (Mabesa and Novero, 1985). Mung bean cereal combinations are the basis of "nutripaks", nutritious food packets for children. These are also useful as weaning foods (Payumo, 1977). Mung bean flour is also the material used for the production of a popular translucent noodle called sotanghon. Mung bean is also widely consumed in other Southeast Asian countries and in East Africa (Kay, 1979). The Institute of Plant Breeding has a strong breeding program for mung bean and has released several widely accepted green and yellow seeded-varieties. Because of the economic significance of this legume and its being a good source of protein in the Filipino diet, our laboratory has undertaken extensive work on its nutritional quality and acceptability and factors affecting them. This paper aims to review our work in the last five years on this topic as well as those of others which are relevant to the topic.

According Sethakorn (2018) to The bean spread was prepared by soaking beans for 1 hour, and then boiled with 1 % of salt with hot water at 95-98° (for 10 minutes using the ratio 1 :2. Bean was left for 15-20 minutes before blending for 1-2 seconds and then it was added 37.5% sugar (w/w). Bean spread was varied by using white bean, mung bean, and peanut to find the suitable type of bean for bean spread. As the results, mung bean showed the highest scores in color, beany flavor, sweetness, and spread ability that are 7.06 ± 1.55 , 7.06 ± 1.46 , 7.00 ± 1.66 , 7.30 ± 1.49 , and 6.86 ± 1.43 , respectively. Mung bean spread was re-produced by adding Matcha powder at 0.5, 1, and 1.5% (w/w). Sensory evaluation results showed the highest preference score at 7.30 ± 0.95 for 1.5% of Matcha powder. As there are the beany flavor affected the flavor of Matcha bean spread, the beany flavor was removed by soaking in 0.8% NaHCO₃ during preparation step. After the evaluation of JAR test, the results showed 60% of panelists satisfied at 60%. Thus, the formula of Matcha green bean spread was mung bean spread 71.96%, sugar 26.54%, and Matcha powder 1.5%. Consumer acceptance test showed the 99% of consumer accept Matcha green bean spread with price of 81 to 90 Baht for 46%. The evaluation scores of color, spread ability, beany flavor, Matcha aroma, sweetness and overall liking were 6.93 ± 1.46 , 7.36 ± 1.09 , 7.18 ± 1.05 , 6.96 ± 1.20 , 7.11 ± 1.11 , and 7.24 ± 0.95 , respectively.

Chumkaew and Punfujinda (2019) Stated in the study that Thai layered dessert or KhanomChanisone of the ancient Thai desserts. This dessert is subtly sweet and slightly oily. The texture is smooth yet sticky. The basic ingredients of Thai layered dessert include coconut milk, sugar,

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and 3-4 other types of flour. The different types of flour used in this dessert give different textures and properties. Tapioca flour makes it soft, sticky, viscous, and transparent. Arrowroot flour hardens it and makes it smooth and sticky, yet less transparent than with tapioca flour. Rice flour hardens it and makes it able to hold its shape easily. Finally, mung bean flour enables this dessert to hold its shape easily and not be too sticky. Two popular ways to eat Thai layered dessert are either eating bit by bit or layer by layer. There are no exact rules regarding how to eat Thai layered dessert but eating it layer by layer gives a sense of sticky, soft, and delicious enjoyment.

According to Poppe (1992) For more than 2000 years connective tissues and products extracted from them have been used in the home and in the food industry for their gelling properties, and also as technical products in the form of adhesives. Availability and product consistency improved when industrial-scale gelatin manufacture appeared at the end of the nineteenth century. More recently, an improved knowledge of amino acids and proteins in general, and of collagen and gelatin in particular, together with the introduction of modern production techniques, has made it possible to produce gelatin which is bacteriologically safe and in accordance with international standards and rigid specifications. The main areas in which gelatin is used, are Food industry (confectionery, meat products, dairy products, etc.), Pharmaceuticals (capsules, etc.), Photography and Technical applications.

Gelatin has been widely used in the food industry as a gelling agent, food stabilizer, emulsifier and thickener. Originating mainly from animals' bones and skin, gelatin is a water-soluble macromolecule with a high molecular weight composed of a mixture of heterologous polypeptides. The gelatin is converted from triple helix collagen after thermal hydrolysis due to the denaturation process which causes alterations to the molecular compositions of amino acid residues (Duconseille et al., 2015). Mammalian gelatin has been widely used in the industry. However, the demand for an alternative source of gelatin has increased due to health and religious concerns. A study showed that gelatin from fish skin could substitute mammalian gelatin (Hosseini & Gómez-guillén, 2018). However, fish gelatin has weaker properties compared with bovine and porcine gelatin (Karim & Bhat, 2009). A Chemical treatment is often used to overcome the problem (Tabarestani et al., 2014). However, gelatin extract usually has high chemical residues (da Silva & Pinto, 2012).

According to Baziwane& He (2003) of all the hydrocolloids in use today, surely none has proven as popular with the general public and found favor in as wide a range of food products as gelatin. A sparkling, clear dessert jelly has become the archetypal gel and the clean melt- in- the- mouth texture is a characteristic that has yet to be duplicated by any polysaccharide. Despite its apparently unfashionable status, more gelatin is sold to the food industry than any other gelling agent. It is relatively cheap to produce in quantity, and there is a ready supply of suitable raw material. The traditional sources of gelatin include bovine and pig skins and demineralized bones and hooves. However, recent studies have shown that there are viable new sources of gelatin such as marine fish skins and bones. Researchers have further sought to develop gelatin derivatives or modified gelatins like coldwater soluble gelatin, hydrolyzed gelatin and esterified gelatin.

Prior Art

Quick and easy sweet Mung Bean dessert recipe by Rathsotheadarachan Sok (2021). If you are looking for a last-minute dessert recipe, then this is it. Sweet Mung bean dessert doesn't require many ingredients and also very simple to make! The ingredient are 220g mung beans (rinsed and soaked for at least 2 hours, drained), 135g brown sugar, 1.5liters water, 2 pandan leaves (rinsed and knotted), 40g sago/tapioca pearls (soaked for 10 minutes, rinsed, and drained) and 240ml coconut milk. The first procedure is to put mung beans, brown sugar, water, and pandan leaves in a large pot and cook over high heat. Once it's boiled, reduce the heat. Then frequently stir it and allow it to simmer for 30 minutes or until mung bean splits open. Next Add sago/tapioca pearls and coconut milk. Let it simmer for another 8 minutes or until sago pearls turn translucent. Add a little water if the mung bean gets too thick and serve warm or chilled. Both the prior art and the present study used milk but differ in kind. Prior art used coconut milk, whereas the present study uses four types of milk for treatment: evaporated, rice, soya, and coconut milk. The ingredients and sugar were the same in both prior art and the present study. Cheese, heavy cream, vanilla, and gelatin powder, on the other hand, are used in the present study to add flavor.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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The product development studies include product formulation, shelf-life testing, proximate analysis and package development.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The study will use an experimental developmental method of research. An experimental involving a manipulation of condition for the purpose of studying the relative effect various treatment applied to member of different sample (worth, 2016)

In this study the experimental method focuses on the investigation of the different milk to be use in the mung beans gelatin chilled research dessert and its effects.

Developmental research is define as the systematic study of designing development nd evaluating instrumental programs, progress and products that must be meet criteria of internal consistency and effectiveness. The most common types of developmental research involve situation in which the product-development process is analyzed and describe and the final product is evaluated (Richey, 2017).

I this study the developmental method is a base on the production of mung beans chilled gelatin dessert. The study will be using a completely randomize design (CRD). using four treatments of varied milk use making the mung beans chilled gelatin dessert.

The study will be using Analysis of Variance (ANOVA) set at 0.01 alpha level and analysis and interpretation of data is the mean. The Nine point Hedonic Scale Instruments will be use in evaluating the sensory attribute during formulation of the product and the acceptability.

Ingredients of Treatment A Puree Mung Beans of 60g, milk evaporated with 80ml, sugar 15g, heavy cream 20g, vanilla 5ml, gelatin powder with 5g, a 500ml of water and 5g of grated cheese.

Ingredients of Treatment B Puree Mung Beans of 60g, milk rice with 80ml, sugar 15g, heavy cream 20g, vanilla 5ml, gelatin powder with 5g, a 500ml of water and 5g of grated cheese.

Ingredients of Treatment C Puree Mung Beans of 60g, milk soya with 80ml, sugar 15g, heavy cream 20g, vanilla 5ml, gelatin powder with 5g, a 500ml of water and 5g of grated cheese.

Ingredients of Treatment D Puree Mung Beans of 60g, milk coconut with 80ml, sugar 15g, heavy cream 20g, vanilla 5ml, gelatin powder with 5g, a 500ml of water and 5g of grated cheese.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objective	Days																													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1. Preparation and experimental of the product																														
2. Sensory Evaluation and consumer testing of the product																														
3. Analysis of data gathered from evaluation																														

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The expected output of this study will be the Development of Mung Beans Gelatin Desserts.

Experimental Procedure

Step 1. Preparation, Gathering Tools, and Ingredients

Prepare and gather tools and equipment needed and ingredients. Measure the ingredients

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accurately according to the exact measurement given.

Step 2. Cooking of Mung Beans

Preparation of the sauce pot on the stove pour 500ml of water and the mung beans will tender and gelatinize. Remove mung beans and drain liquid let the mung beans cool. Pour cool mung beans into a blender and blend well until they become puree in texture. Set aside.

Step 3. Cooking and Mixing Mung Beans and other milk and other ingredients

Combine the other ingredients such as milk, sugar, heavy cream, and vanilla then add the mung bean puree. Then mix well until the mung beans all mixture is combined. set aside.

Step 4. Preparation of Powder Gelatin

Boil 1/2 cup of water, then measure the gelatin powder into it. measure 2 tablespoons of warm water dilute it with gelatin powder and add gelatin mixture into the mung beans mixture. Pour the mung beans mixture into a molder container and let it chill at least overnight. Put grated cheese on top.

In one mixture produce there will be approximately three (3) pieces with 2.5 oz cup to chill and serve.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

To produce recipe and cooking guide materials and brochures will be disseminated and distributed to various interested individuals.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this study will be the following: chicken grower, noodle makers or producers, vegetable growers, homemakers, children, food enthusiast, and sausage vendors.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Juvy I. Dordas	Project leader	Foods

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Sources of Expenses	Quantity	Amount
A. Ingredients (In Four treatments)		
Mung Beans	3 packs (180g)	160
Milk :		
Rice Milk	1 box	325
Soya Milk	1 bottle	105
Heavy cream	1 pcs	180
Sugar	3 box	70
Vanilla	1 kilo	45
Gelatin Power	1 small bottle	180
Cheese	1 box	60

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Container Cap	1 box 12pcs	60	
B. Microbial and proximal Analysis, and Shelf-life test		5,000.00	
C. Miscellaneous		1,000.00	
Total Expenses		Php 7,350.00	

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)

JUVY I. DORDAS, was born on March 27, 1975 at Roxas City, Capiz. She spent her elementary years at Plaridel East Elementary School, and her secondary education at Capiz Institute of Technology Roxas City. She obtained her bachelor's degree at Capiz Institute of Technology Roxas City with the degree of Bachelor of Science in Industrial Education major in Home Economics. She finished her master's and doctorate degree at the same university. Presently, she is an Assistant Professor IV handling major subject (Food Technology under Industrial Technology Department)